

MIP_V01_EXECUTION_PLAN.md

Mainframe Intelligence Platform

Phase 1 - Discovery Foundation

Version: 1.0

Goal

Build the first working version of MIP.

The platform should answer:

1. What programs exist?
2. What jobs exist?
3. What copybooks exist?
4. Which jobs execute which programs?

Nothing else.

Scope

In Scope:

- Repository Discovery
- COBOL Catalog
- JCL Catalog
- SQLite Metadata Store
- NetworkX Relationship Graph

Out of Scope:

- AI
 - Business Rules
 - Impact Analysis
 - Modernization Intelligence
 - Service Discovery
-

Technology Stack

Python 3.13

FastAPI

Pydantic v2

SQLite

NetworkX

Google ADK

Pytest

Ruff

Workspace Structure

mip-platform/

```
|—— mfcode/ | |—— src/ | |—— domain/ | |—— discovery/ | |—— parsers/ | |——  
repository/ | |—— graph/ | |—— api/ | |—— common/ | |—— tests/ | |—— output/ |  
|—— docs/ | |—— skills/ | |—— prompts/
```

Deliverable 1

Repository Scanner

Purpose:

Discover all artifacts.

Output:

artifact_inventory.json

Example:

```
{ "type": "cobol", "path": "cobol/PAY001.cbl" }
```

Deliverable 2

Program Catalog

Extract:

PROGRAM-ID

Example:

PROGRAM-ID. PAY001.

Output:

program_catalog.json

Deliverable 3

Job Catalog

Extract:

JOB

EXEC PGM

Output:

job_catalog.json

Deliverable 4

SQLite Metadata Store

Tables:

artifacts

programs

jobs

job_program_relationships

Deliverable 5

NetworkX Graph

Nodes:

Program

Job

Relationships:

EXECUTES

Deliverable 6

First Query

Question:

Which jobs execute PAY001?

Success Criteria

User can point MIP to a repository and answer:

Which jobs execute a program?

Without manually reading source code.

Completion Checklist

Repository Scanner Complete

COBOL Catalog Complete

JCL Catalog Complete

SQLite Complete

NetworkX Complete

Query API Complete

Unit Tests Complete

Documentation Complete

Phase 1 Complete